```
#include <iostream>
#include <climits>
using namespace std;

int main()
{
   int n;
   int cost[10][10];

   cout << "Enter number of vertices: ";
   cin >> n;

   cout << "Enter cost matrix:\n";
   cout << "Enter 0 for no direct path.\n";

   for (int i = 0; i < n; i++)
   {
      for (int j = 0; j < n; j++)
      {
         cin >> cost[i][j];

         if (i != j && cost[i][j] == 0)
         {
            cost[i][j] = INT_MAX;
         }
      }
   }

   // Floyd-Warshall Algorithm
   for (int k = 0; k < n; k++)
   {
```

```
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            if (cost[i][k] != INT_MAX &&
                cost[k][j] != INT_MAX)
            {
                if (cost[i][j] > cost[i][k] + cost[k][j])
                {
                    cost[i][j] = cost[i][k] + cost[k][j];
                }
            }
        }
    }
}

cout << "\nMinimum Cost Matrix:\n";

for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (cost[i][j] == INT_MAX)
            cout << "INF\t";
        else
            cout << cost[i][j] << "\t";
    }

    cout << endl;
}
```

```
    return 0;
}
```

```
Enter number of vertices: 2
Enter cost matrix:
Enter 0 for no direct path.
0
1
3
4

Minimum Cost Matrix:
0       1
3       4
```